

When Should an Alert Threshold Learn?

Admission-separated recalibration as a decision-time action for time-series anomaly alerting

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A threshold that learns only from “normal” windows can only fail

A threshold keeps its last 3 admitted scores, sets itself to their maximum, and admits a new score **only if it judged that score normal**. Score 9 is normal at step 1 and an alert at step 6: the alerts come from the threshold, not the signal. Admitting *every* score fails the other way, because a sustained fault becomes the new normal.

Step	Score	Judged against	Decision	Threshold after
1	9	10	normal, admitted	9
2	12	9	alert, rejected	9
3	8	9	normal, admitted	9
4	7	9	normal, admitted	9
5	6	9	normal, admitted	8
6	9	8	alert , rejected	8

Our direction: recalibration as a guarded, auditable action

Recalibrate only as a deliberate decision, from data chosen **neither by the threshold itself nor unconditionally**, and judge every option on **one common score trace**.

Evidence

What we claim

PROVED

C1: learning only from “normal” windows means the threshold never rises

MEASURED

C2: on SKAB that costs $16.2\times$ and $3.9\times$ the alert windows

MEASURED

C3: the scorer, not the policy, limits detection

IMPLEMENTED

C4: a four-action controller, recalibration bounded to $[\theta_0, 4\theta_0]$

MEASURED

C5: identical to fixed on SKAB, so no harm and no benefit

MEASURED

C6: under stress it works on a step and fails on a ramp

PROPOSED

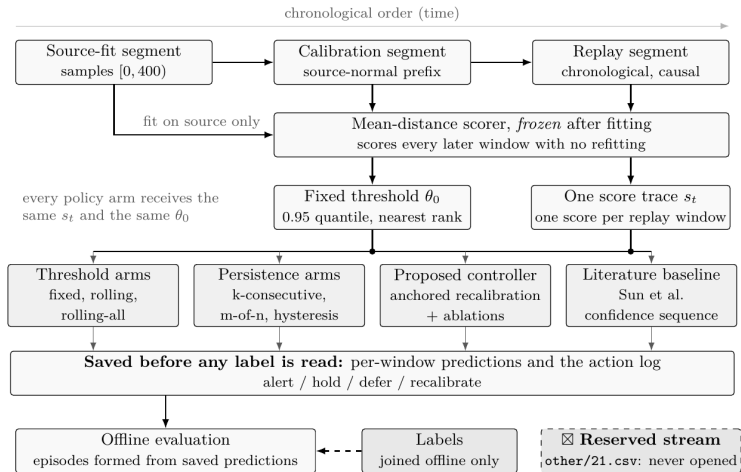
C7: a benefit on real regime changes, decided by a kill test

Isn't this just X? Each ingredient exists; the combination does not

Open in the nine papers checked: action policies compared on one frozen score trace. A bounded search, not proof of absence.

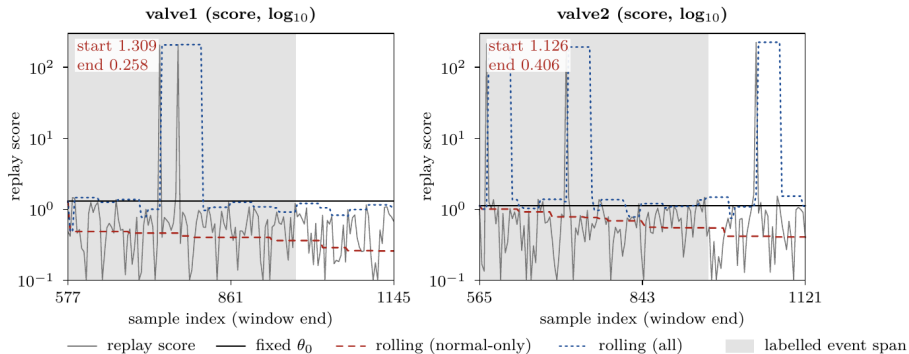
Closest work	Already does	Ours differs
FITNESS (ICML 2022)	Shows dropping self-flagged anomalies fails	We claim only the ratchet proof
Sun et al. (ICML 2024)	Abstains with guarantees	Re-implemented and measured
Perini & Davis	Rejects on a fixed stability threshold	Defer is charged to coverage
DDADE, ADAPTS, CDDIA	Retrain the model on drift	Scorer frozen; only actions change
Hu (in press)	Evaluates alerts by workload	Same frame, plus a controller
SEAD, Dynamic-XY	New score; suppression from alert history	One score trace, explicit actions

Every policy sees identical scores; labels arrive only afterwards



Proved, then observed: the self-referential threshold only falls

PROVED **Proposition 1.** Admit s only if $s \leq \theta$ and set θ to the buffer's nearest-rank quantile; then $\theta_{t+1} \leq \theta_t$ for every score sequence. **MEASURED** **0** increases in 143 and 140 steps; valve1 1.309 $\rightarrow$ **0.258**, valve2 1.126 $\rightarrow$ **0.406**.



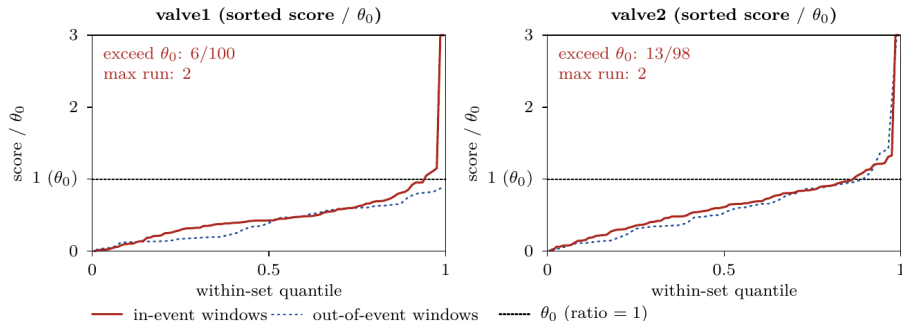
Same single event, up to 16× the alerts; our controller equals fixed

MEASURED 97 vs 6 (16.2×) and 70 vs 18 (3.9×) alert windows for the same event. Our controller: 0 deferrals, 0 recalibrations, **no harm and no evidence of benefit**. *False-alert time* = alerted samples outside the labelled event.

Arm (valve1 / valve2)	Alert windows	False-alert time	Recall	Coverage
Fixed threshold	6 / 18	0 / 20	1/1	full
Rolling, normal-only	97 / 70	109 / 104	1/1	full
Rolling, admit-all	10 / 13	10 / 12	1/1	full
k-consecutive	0 / 0	0 / 0	0/1	full
Our controller	6 / 18	0 / 20	1/1	full
Sun et al.	3 / 2	0 / 4	1/1	54/143 · 57/140

The scorer, not the policy, limits detection on SKAB

MEASURED Only **6/100** (valve1) and **13/98** (valve2) in-event windows exceed θ_0 , never more than **2** in a row, so rules needing 3 miss the events. Each event spans about **70%** of replay, so recall 1/1 says little. **A policy can move alerts in time; it cannot create separation the scorer lacks.**

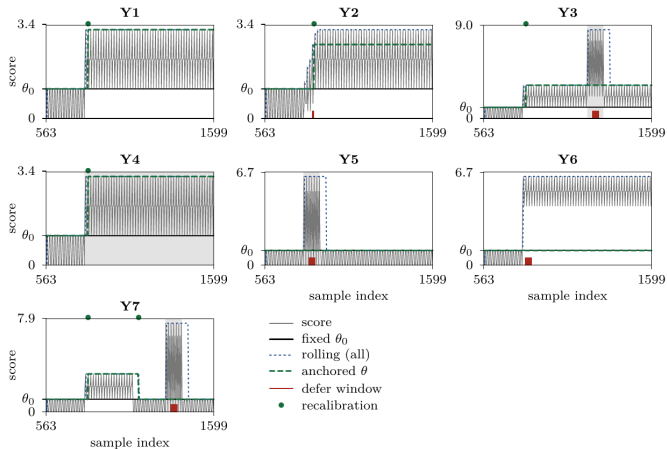


The controller: four actions, recalibration only inside a bounded band

IMPLEMENTED AnchoredRecalibrationPolicy. Its buffer holds the last 10 scores of **every** window, so what it learns from never depends on its own decisions. Nothing was tuned: 10 and 0.95 are the rolling baseline's values; the cap 4 and the stability bound 2 are declared operating assumptions.

At a window end	Action
No sustained rise	hold , or alert if $s > \theta$
Sustained rise that is stable and at most $4\theta_0$	recalibrate up to it
Sustained rise, guard fails	defer , at most 10 in a row
Still failing after 10 deferrals	escalate : alert-capable, no recalibration
Rise over	recalibrate back toward θ_0

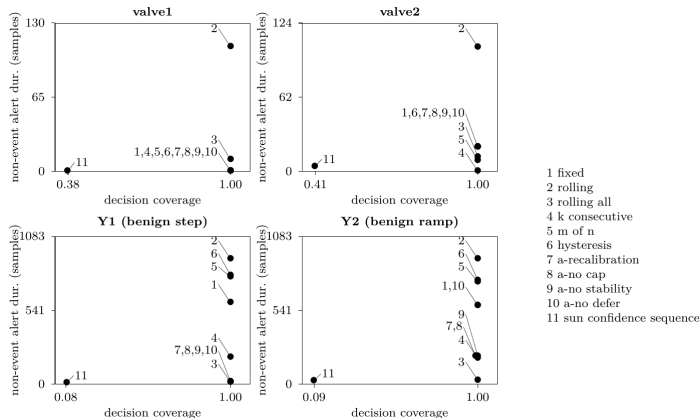
Under stress: adapts to a step, fails on a ramp and a step-shaped fault



MEASURED Seven deterministic cases:

- **Works, Y1 step:** false-alert time **16** vs fixed 597, coverage 260/260
- **Still detects, Y3 and Y7:** later faults caught, recall 1/1
- **Fails, Y2 ramp:** locks into alerts, **204** over 50 episodes
- **Fails as declared, Y4:** a fault shaped like a step is absorbed, 0.020 vs fixed 0.746

Quiet has two routes: adapt while deciding, or stop deciding



How to read it: one marker per policy; further right decides on more windows, lower means less false-alert time.

MEASURED On Y1 our controller reaches **16** while deciding on **260/260** windows.

Sun et al. are quiet because they decide on only **22/260**.

The strongest case against us: a simpler rule is quieter

These results do not establish a useful controller.

- **Admit-all is quieter on benign shifts:** false-alert time 12 vs our 16 on Y1, and 28 vs our 204 on Y2.
- **Our only edge is keeping faults visible:** on Y3 it alerts 0.29 of the fault against admit-all's 0.04, both at recall 1/1.
- **SKAB gives no evidence either way:** our rows are identical to fixed.
- **Abstention buys quiet with coverage:** Sun et al. decide on 54/143 and 57/140 windows.

What we claim instead: a proved mechanism, an honest negative-to-inconclusive development finding, and a test that can refute the direction.

What Phase-I shows, and what it does not

Shown

- **PROVED** the ratchet, for any score sequence
- **MEASURED** its cost on two SKAB streams
- **MEASURED** the scorer limit and recall saturation
- **IMPLEMENTED** a tested four-action controller
- **MEASURED** no harm on SKAB; a mapped failure set

Not shown

- any benefit on real benign regime changes
- any held-out or generalisation result
- anything about `other/21.csv`: reserved, never opened

Limits: one event per stream, one channel (Current), dataset licence not independently verified.

The next experiment is designed to be able to kill the idea

PROPOSED Replay all arms on common traces over every other SKAB valve1 and valve2 file, admitted by a label-independent rule, plus one public family with labelled **benign** regime changes.

The direction is refuted if any one holds:

1. Where the elevation test fires, it does **not** cut false-alert time versus fixed at equal recall.
2. It **loses** an event that fixed detects.
3. Sun et al.'s abstention **matches** it at equal coverage.

A refutation becomes the reported result.